THE HILBERT-SAMUEL FUNCTION OF A FINITE MODULE AND OF A FINITE-LENGTH QUOTIENT DIFFER BY A BOUNDED AMOUNT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finite length of a quotient gives a power of the ideal inside the submodule. Used at the first line of the counted proof (line 14273). Supplies the constant c_2 with I^{c_2}M contained in M'.
- Additivity of length over short exact sequences. Used repeatedly in the two displayed computations (lines 14279-14299). Splits the length of M modulo a power of I into the length of the quotient M/M' plus a length computed inside M'.

Lemma 0.1. *Suppose that $M' \subset M$ are finite R-modules with finite length quotient. Then there exists a constants c_1, c_2 such that for all $n \geq c_2$ we have*

$$c_1 + \chi_{I,M'}(n - c_2) \leq \chi_{I,M}(n) \leq c_1 + \chi_{I,M'}(n)$$

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Proof. Since M/M' has finite length there is a $c_2 \geq 0$ such that $I^{c_2}M \subset M'$. Let $c_1 = \text{length}_R(M/M')$. For $n \geq c_2$ we have

$$\begin{aligned} \chi_{I,M}(n) &= \text{length}_R(M/I^{n+1}M) \\ &= c_1 + \text{length}_R(M'/I^{n+1}M) \\ &\leq c_1 + \text{length}_R(M'/I^{n+1}M') \\ &= c_1 + \chi_{I,M'}(n) \end{aligned}$$

On the other hand, since $I^{c_2}M \subset M'$, we have $I^nM \subset I^{n-c_2}M'$ for $n \geq c_2$. Thus for $n \geq c_2$ we get

$$\begin{aligned} \chi_{I,M}(n) &= \text{length}_R(M/I^{n+1}M) \\ &= c_1 + \text{length}_R(M'/I^{n+1}M) \\ &\geq c_1 + \text{length}_R(M'/I^{n+1-c_2}M') \\ &= c_1 + \chi_{I,M'}(n - c_2) \end{aligned}$$

which finishes the proof. □